

Permutation Mastermind in $O(n)$ Expected Queries

Communicated by Bernardo Subercaseaux. Proof from AIs

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Abstract

In permutation Mastermind, both the secret and every query are permutations of $[n]$, and a query is answered by its number of agreements with the secret. We give an adaptive randomized strategy that identifies the secret correctly, uses $O(n)$ queries in expectation against every fixed secret, and never uses more than $O(n \log n)$ queries. The expected bound is optimal up to an absolute constant factor.

1 Introduction

In permutation Mastermind, also called the AB game, a codemaker chooses a permutation σ^* of $[n] = \{1, \dots, n\}$. A codebreaker proposes a permutation q and learns only the number of positions at which $q(i) = \sigma^*(i)$. Each query may depend on the answers to earlier queries. How many queries are needed to identify the secret?

Theorem 1.1. *For every $n \geq 2$, there is an adaptive randomized strategy for permutation Mastermind that terminates with the correct secret on every execution. Its expected number of queries is $O(n)$, uniformly over secrets, and its number of queries on any execution is $O(n \log n)$.*

El Ouali, Glazik, Sauerland, and Srivastav [EGSS18] give a deterministic strategy using $O(n \log n)$ queries for this game. For ordinary black-peg Mastermind with n positions and n colors, Martinsson and Su [MS24] obtain $O(n)$ expected queries. That model permits repeated colors in both the secret and the queries. Our theorem attains the same expected order when both must be permutations.

We count queries, allowing finite exhaustive computation between them. The main task is the expected bound; a final truncation using the deterministic strategy above supplies the bound on every execution. The expected order is optimal: every always-correct strategy that halts almost surely has a secret for which the expected number of queries is at least $\log(n!)/\log(n+1) = (1 - o(1))n$.¹

It is useful to view permutations as matchings, as in [Sub26]. Let U and V be disjoint copies of $[n]$, representing positions and symbols. The edge (i, c) assigns symbol c to position i . The secret is a perfect matching M^* in $K_{n,n}$, a query is another perfect matching Q , and its score is $|Q \cap M^*|$. The restriction on queries is now explicit: every vertex must be used exactly once.

¹Almost every random tape gives a correct, terminating strategy on all $n!$ secrets. Its terminal answer sequences form a prefix-free code over an alphabet of size $n+1$. By Kraft's inequality their mean length is at least $\log(n!)/\log(n+1)$. Averaging over tapes gives this lower bound for the mean over secrets, and hence for some secret.

A score counts agreements without locating them. Thus “keep querying until the first position is correct” is not a stopping rule the codebreaker can follow. Every stopping condition in our strategy must be recognizable from the queries, their answers, and the strategy’s own choices. Score zero is useful precisely because it provides such a certificate: every edge of the query is false.

1.1 The proof strategy

Fix a remaining board of size N . It is enough to identify a fixed positive fraction of its secret edges using $O(N)$ expected queries: repeating this step on the remaining board gives a geometric sum. We call this step a *phase*.

We first make partial matchings measurable. A partial matching must be completed before it can be queried, and the added edges may change its score. Queries of score zero supply edges whose contribution is known to be zero. After collecting enough of them, a matching argument lets us complete every partial matching of the sizes we shall use without changing its score.

We then construct linearly many edge-disjoint partial matchings of score one. Each contains a single secret edge, which could be found by binary search. Searching them separately would cost $O(N \log N)$ queries. The remaining task is to combine these searches without paying for every binary test individually.

There are two obstacles. The subsets tested by different searches may share vertices, so their union need not be a matching. Even when the union is measurable, its score gives only the sum of the test answers, not the answers themselves. We must arrange both that the tests fit together and that their sums retain enough information.

To control the first obstacle, we use common nested partitions of the two vertex sets and alternately restrict the possible endpoints of each secret edge. We organize the searches into $O(\log N)$ *epochs*, with each active search advancing one step per epoch. Linear total cost calls for $O(N / \log N)$ measurements per epoch. We stagger the starting times so that only $O(N / \log N)$ searches are at any one depth: a new group begins while earlier groups continue at smaller scales.

To recover the separate answers, we place repeated copies of the tests into compatible collections and measure signed sums. We prove that these sums can be chosen to determine every test answer using $O(N / \log N)$ pairs of measurements per epoch. Deeper tests receive more copies, but their smaller supports make those copies less costly to accommodate. Making these geometric and informational requirements fit together is the main part of the proof.

The strategy verifies the encoding before using it. A failed search for an encoding causes a restart, never an incorrect answer. The probability estimates bound the cost of these restarts. Since there are $O(\log N)$ epochs, the expected cost of a phase is $O(N)$. We state the detecting theorem when the encoding is needed, use it to complete the strategy in [Section 6](#), and then prove it in [Section 7](#).

1.2 Conventions

Deleting the endpoints of identified secret edges leaves a *residual* board $K_{N,N}$, with secret still denoted M^* . To evaluate a residual query, adjoin the identified edges and subtract their number from the answer. From [Section 2](#) onwards, queries and scores refer to the residual board.

All constants are absolute unless a dependence is stated. We choose a cutoff N_0 after the other constants, and work with $N \geq N_0$. Smaller boards are solved by querying their permutations in turn, at bounded cost. A randomized procedure is *zero-error* if it halts almost surely and always gives the correct answer; the final strategy will also terminate on every execution. We regard a uniform draw from a finite set as one random step; an execution is a sequence of these draws. We write $\log$ for the natural logarithm and $\log_2$ for logarithms to base two.

2 Making partial matchings measurable

Using $O(N)$ expected queries, we shall arrange that every partial matching of size at most $7N/8$ can subsequently be measured with one query. The problem is to fill its uncovered vertices without adding an unknown contribution to the score.

Call an edge *certified false* if it has appeared in a query of score zero. Write F for the set of these edges and $G = K_{N,N} \setminus F$ for the *candidate graph*. Thus $M^* \subseteq G$. A small maximum degree in G will give us enough false edges for the required completions.

Lemma 2.1 (False-edge reservoir). *For every $N \geq N_0$ and every residual secret, a procedure using $O(N)$ expected queries produces a set F of certified-false edges whose complement satisfies $\Delta(G) \leq N/16$.*

Proof. Query independent uniformly random permutations and retain those of score zero. The retained queries are independent and uniform among the score-zero queries; by symmetry, each uses every false neighbor of a fixed vertex with probability $1/(N-1)$.

Group the retained queries into batches of size $16N$, and put $s = \lfloor N/16 \rfloor$. If a vertex still has candidate degree greater than $N/16$ after a batch, at least s of its false neighbors remain unseen: its degree is at least $s+1$, and only one incident edge is secret. Thus some set of s false neighbors was avoided throughout the batch. A union bound over these sets and all vertices gives

$$\begin{aligned} \mathbb{P}(\Delta(G) > N/16 \text{ after the batch}) &\leq 2N \binom{N-1}{s} \left(1 - \frac{s}{N-1}\right)^{16N} \\ &\leq 2N \binom{N-1}{s} e^{-16s} \leq Ne^{16} \left(\frac{2}{e}\right)^N = o(1). \end{aligned} \tag{1}$$

The last inequality uses only $\binom{N-1}{s} \leq 2^{N-1}$ and $16s \geq N-16$: the avoidance probability pays for all subsets of neighbors.

Repeat fresh batches until the known graph satisfies $\Delta(G) \leq N/16$. For sufficiently large N_0 , a fresh batch suffices with probability at least $1/2$, regardless of the preceding history, so the expected number of batches is at most two. A random query has score zero with probability $p_0 = \sum_{j=0}^N (-1)^j / j! \geq 1/3$ for $N \geq 2$, by inclusion–exclusion and the alternating-series bounds. Each batch has expected cost $16N/p_0$. Wald’s lemma, applied to the successive independent batch experiments and the stopping time determined by their records, gives expected total cost at most $32N/p_0 = O(N)$. $\square$

The fraction $1/16$ is convenient rather than essential. The same argument can leave any prescribed small fixed fraction of the edges at each vertex uncertified. We need not certify every false edge.

The remaining question is whether the unused vertices can be matched using only false edges. We use the following fact.

Lemma 2.2 (A matching in a dense bipartite graph). *A bipartite graph with r vertices on each side and minimum degree at least $r/2$ has a perfect matching.*

Proof. Choose a maximum matching $\mathcal{M}$. If it is not perfect, let u and v be unmatched vertices on opposite sides. All their neighbors are matched. The neighbors of u meet $\deg(u)$ edges of $\mathcal{M}$, and those of v meet $\deg(v)$ edges. Since

$$\deg(u) + \deg(v) \geq r > |\mathcal{M}|,$$

some edge $u'v' \in \mathcal{M}$ is met by both sets of neighbors. Replacing $u'v'$ by uv' and $u'v$ enlarges the matching, contradicting its choice as a maximum matching. $\square$

Lemma 2.3 (Exact completion). *Suppose $\Delta(G) \leq k$. Every partial matching E with $|E| \leq N - 2k$ extends to a perfect matching whose added edges lie in F . Consequently, one query determines $|E \cap M^*|$.*

Proof. Delete the endpoints of E and write $r = N - |E| \geq 2k$. On the remaining r -by- r board, each vertex has at most k neighbors in G and hence at least $r - k \geq r/2$ neighbors in F . By Lemma 2.2, the remaining F -edges contain a perfect matching. Adjoin it to E . Its edges contribute zero to the score, so the completed query measures E (Fig. 1). $\square$

We call this determination of $|E \cap M^*|$ a *measurement* of E . The completion can be found by finite search, since F is known. With $k \leq N/16$, every partial matching of size at most $7N/8 \leq N - 2k$ can be measured with one query.

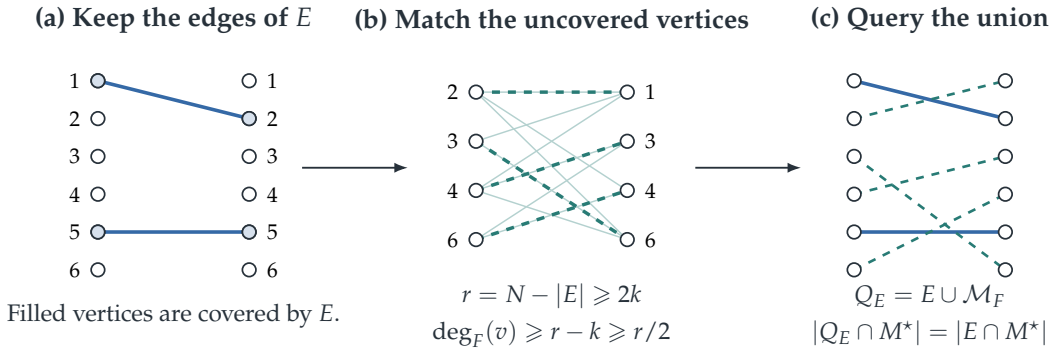


Figure 1: A partial matching E is completed using certified-false edges. Pale lines show available F -edges; dashed lines form the added matching. They contribute zero to the score, so the completed query measures E .

3 Isolating a linear number of secret edges

A partial matching of score one contains a single secret edge. Every subset has score zero or one, so finding that edge becomes a sequence of binary questions. We shall first obtain many such partial matchings, then arrange to search them together.

Fix the state reached by Lemma 2.1, and put $k = \Delta(G) \leq N/16$. Keep F and G fixed for the rest of the phase, even if later queries certify more false edges. A *cell* is a partial matching certified to contain exactly one secret edge. We call this its *marked edge*, and its endpoints the *marked endpoints*.

Lemma 3.1 (Score-one cells). *With F , G , and k fixed as above, $O(N)$ expected queries produce pairwise edge-disjoint cells $Q_1, \dots, Q_L$ such that*

$$L \geq \lambda_0 N, \quad |Q_j| \leq \lceil N/2 \rceil \quad (1 \leq j \leq L).$$

One may take $\lambda_0 = 1/(4e)$. The marked edges are distinct, and their endpoints are distinct on each vertex side.

Proof. König's line-coloring theorem partitions $E(G)$ into k matchings $C_1, \dots, C_k$. Split each into at most two pieces of size at most $\lceil N/2 \rceil$ and measure them. This costs at most $2k \leq N/8$ queries and determines $h_j = |C_j \cap M^*|$. Since the classes partition the candidate graph,

$$\sum_{j=1}^k h_j = N. \quad (2)$$

In one *round*, process each class with $h = h_j > 0$ as follows. Independently label every edge uniformly from $[h]$, partitioning the class into h buckets. Split each nonempty bucket into at most two pieces of size at most $\lceil N/2 \rceil$, and measure every piece. For $h = 1$ this is simply a one-bucket partition. The size bound makes all measurements legal, and a round costs at most

$$2 \sum_{j=1}^k h_j = 2N \quad (3)$$

queries. Let Y be its number of score-one pieces.

A secret edge is *isolated* in its bucket if no other secret edge receives its label. In a class with $h \geq 2$ secret edges, a given one is isolated with probability

$$\left(1 - \frac{1}{h}\right)^{h-1} \geq e^{-1}; \quad (4)$$

the inequality follows from $(1 + 1/(h-1))^{h-1} \leq e$. For $h = 1$ the probability is one. Each isolated edge produces a distinct score-one piece, so $\mathbb{E}Y \geq N/e$. Also $Y \leq N$: the pieces are edge-disjoint, and each score-one piece contains a different secret edge. If $p = \mathbb{P}(Y \geq N/(2e))$, then

$$N/e \leq \mathbb{E}Y \leq N/(2e) + Np, \quad \text{hence } p \geq 1/(2e). \quad (5)$$

A score-one piece need not arise from an isolated bucket. Isolation is only a sufficient condition used in the analysis (Fig. 2).

Accept the round when $Y \geq L_0 := \lfloor N/(2e) \rfloor$; otherwise discard its pieces and repeat with fresh labels. The number of rounds is dominated by a geometric random variable of success probability $1/(2e)$. Thus the rounds cost $O(N)$ queries in expectation. Including the initial $2k$ class measurements does not change this bound.

Retain any $L = L_0$ score-one pieces from an accepted round. For sufficiently large N , $L_0 \geq N/(4e)$. These cells are pairwise edge-disjoint, so their marked edges are distinct. Since the marked edges lie in M^* , their endpoints are distinct on both sides. $\square$

We could now find every marked edge by searching its cell separately. Each binary question asks whether the edge lies in a specified subset of the cell; call the question a *test*, and that subset its *support*. Since each cell has at most N edges, separate binary searches use $O(N \log N)$ queries. From this point on, our task is to avoid paying separately for all these tests.

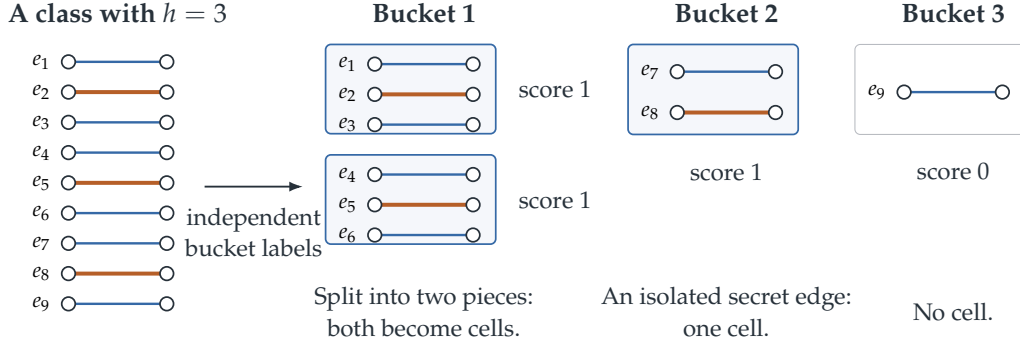


Figure 2: A bucket containing two secret edges can split into two score-one pieces. Thus score-one pieces can outnumber isolated secret edges. When secret edges are drawn, orange edges or circles display them for the reader; this information is not given to the strategy.

4 Searching at several depths at once

The cells are edge-disjoint, but they may share vertices. Their test supports therefore need not unite into a partial matching. To combine the searches, we shall confine their supports to rectangles that shrink on both vertex sides, and arrange that only a few searches occupy any one scale at a time.

4.1 One search

Choose an order of each vertex set. In each order, divide the vertices into consecutive blocks, then divide each block into two nearly equal children, and continue until every nonempty block is a singleton. Every cell uses the same two *hierarchies*. Thus blocks from different searches are comparable: on either side, two blocks are disjoint or one contains the other.

Within a cell, either endpoint determines the marked edge, since the cell is a matching. Nevertheless, we search for both endpoints. Restricting the left endpoint makes a test smaller, but need not confine its right endpoints to a single small block. Restricting both endpoints supplies the geometric control needed to combine tests from different cells.

Here is the precise rounding convention. Put $d = \lceil \log_2 N \rceil$, and write either vertex order as $v_1, \dots, v_N$. Its blocks at depth r are

$$B_{r,b} = \left\{ v_i : \lfloor \frac{bN}{2^r} \rfloor < i \leq \lfloor \frac{(b+1)N}{2^r} \rfloor \right\}, \quad 0 \leq b < 2^r. \quad (6)$$

A block has children $B_{r+1,2b}$ and $B_{r+1,2b+1}$, and size at most $\lceil N/2^r \rceil$. Empty blocks are allowed. At depth d , every nonempty block is a singleton. When N is a power of two, these are the usual dyadic intervals in the chosen order. We will draw the two orders at random after specifying which cells begin their searches together.

A cell Q is a fixed partial matching, and its marked edge is fixed but unknown. What changes during the search is its *frame* $P \times H$, where P and H are blocks of the left and right hierarchies. The frame initially equals $U \times V$ and always satisfies

$$|Q \cap (P \times H) \cap M^*| = 1. \quad (7)$$

At stage $u = 2r$, split the left block, $P = P^0 \sqcup P^1$, and use the test support

$$S = Q \cap (P^0 \times H). \quad (8)$$

Its score is a bit. An answer of 1 retains P^0 ; an answer of 0 retains P^1 . At stage $u = 2r + 1$, split the right block, $H = H^0 \sqcup H^1$, and use

$$S = Q \cap (P \times H^0) \quad (9)$$

to choose its child in the same way. Each update preserves (7). The support is a subset of the fixed cell inside part of its current frame (Fig. 3).

If one child rectangle has empty intersection with Q , the answer is already determined: the marked edge lies in the other child. We make this *forced* update without a measurement, but still count it as a stage. The invariant rules out both intersections being empty.

At the beginning of stage u , the two block depths are

$$d_U(u) = \lceil u/2 \rceil, \quad d_V(u) = \lfloor u/2 \rfloor. \quad (10)$$

At either stage $2r$ or $2r + 1$, the test restricts one endpoint to a depth- $(r + 1)$ block. Since its support is a matching, it has at most

$$s_r = \lceil N/2^{r+1} \rceil \quad (11)$$

edges. After $D = 2d$ stages, both blocks are singletons and the marked edge is known. To combine tests, we will require their frames to be disjoint on each vertex side. This guarantees disjoint supports, and the common hierarchies make the stronger condition easier to control.

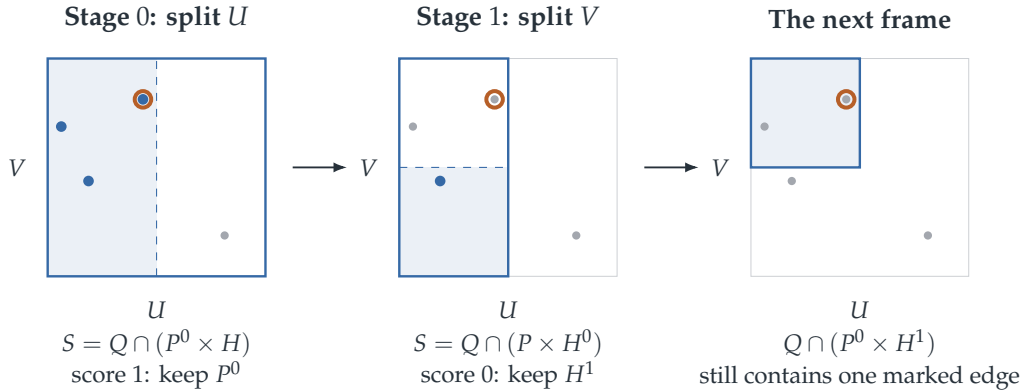


Figure 3: The cell Q is the same matching in all three panels; each dot represents one edge. Its frame (blue boundary) shrinks, and its test support (blue dots) changes. The answers 1 and 0 retain first the left child and then the upper child.

4.2 The schedule

Each search needs $D = O(\log N)$ stages. We shall carry them out over $O(\log N)$ epochs, with each active search advancing one stage per epoch. To distribute a linear query budget over these epochs, we aim for $O(N / \log N)$ measurements in each.

If all searches began together, their first-stage frames would all be $U \times V$. Under our compatibility rule, each non-forced first-stage test would then need a separate collection, or *row*. We therefore admit only $O(N / \log N)$ new searches at a time. Older searches continue alongside them, at progressively smaller scales.

Divide the L cells into groups of approximately L/D cells. More precisely, put

$$d = \lceil \log_2 N \rceil, \quad D = 2d, \quad m = \lfloor L/D \rfloor, \quad J = \lceil L/m \rceil. \quad (12)$$

Since $\lambda_0 N \leq L \leq N$, increasing N_0 ensures $L/D \geq 2$, and hence

$$m \geq \frac{L}{2D}, \quad J \leq 2D + 1. \quad (13)$$

Thus $m = \Theta(N/\log N)$ and $J = O(\log N)$. Partition the cells into J cohorts, each of size at most m . With the cells and this partition fixed, draw the two vertex orders independently and uniformly at random.

Number the cohorts $0, \dots, J-1$. Cohort j begins in epoch j and advances one stage per epoch: in epoch e , it performs stage $u = e - j$ whenever $0 \leq e - j < D$. There are $J + D - 1$ epochs altogether, with at most m tests at each stage in any epoch (Fig. 4). Advancing each cell only once per epoch also respects the dependence on earlier answers: once the preceding epochs have been decoded, every current test is known.

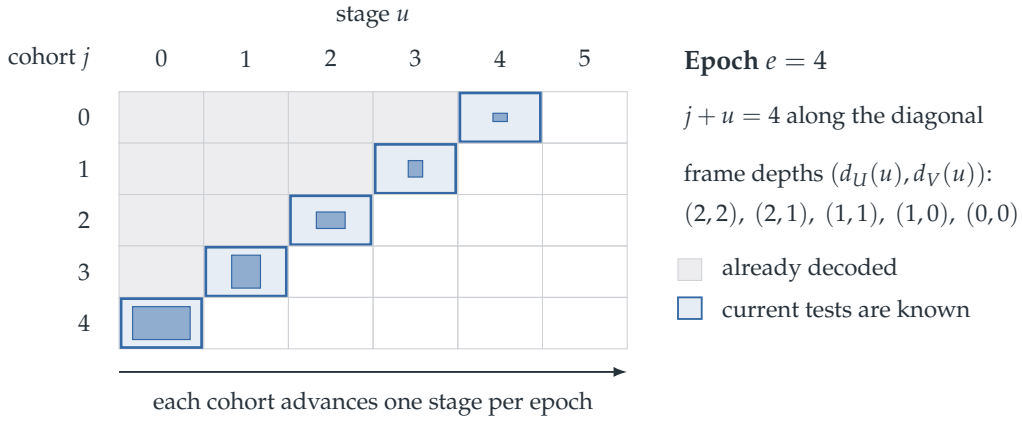


Figure 4: One epoch is a diagonal through the schedule, with at most m tests at each stage. The rectangles indicate relative frame sizes. The grey region has already been decoded, so every test on the highlighted diagonal is known.

4.3 Balance of the hierarchies

Tests at the same stage may still concentrate in a few blocks. Every frame side contains its cell's marked endpoint, so we can control this concentration by counting marked endpoints in blocks. The endpoints are fixed; their block occupancies are random.

Lemma 4.1 (Simultaneous hierarchy balance). *There is an absolute constant C_{bal} such that, with probability at least $1 - N^{-10}$, the following holds for every cohort, every depth $0 \leq r \leq d$, and every depth- r block B on either side:*

$$\#\{\text{marked endpoints of the cohort in } B\} \leq C_{\text{bal}} \left(\frac{m}{2^r} + \log N \right). \quad (14)$$

Proof. Fix the cells, their marked edges, and the cohort partition. The marked endpoints of a cohort are distinct on each side. For a cohort with $h \leq m$ endpoints on one side and a specified depth- r block of size b , their number X_B in the block has a hypergeometric distribution, with mean

$$\mu = \mathbb{E}X_B = \frac{hb}{N} \leq \frac{m}{2^r} + 1. \quad (15)$$

For an integer $1 \leq q \leq h$, a union bound over sets of q marked endpoints gives

$$\begin{aligned} \mathbb{P}(X_B \geq q) &\leq \binom{h}{q} \frac{(b)_q}{(N)_q} \\ &\leq \binom{h}{q} \left(\frac{b}{N}\right)^q \leq \left(\frac{e\mu}{q}\right)^q, \end{aligned} \quad (16)$$

where $(x)_q = x(x-1)\cdots(x-q+1)$. If $q > b$ or $q > h$, the event is impossible.

Take $q = \lceil C_{\text{bal}}(m2^{-r} + \log N) \rceil$. For $N \geq 3$ we have $\mu \leq m2^{-r} + \log N$. Choosing C_{bal} with $C_{\text{bal}} \log(C_{\text{bal}}/e) > 20$ therefore yields

$$\mathbb{P}(X_B > C_{\text{bal}}(m2^{-r} + \log N)) \leq N^{-20}. \quad (17)$$

There are fewer than $4N$ blocks in either hierarchy and $J = O(\log N)$ cohorts. A union bound over both sides, all blocks, and all cohorts proves the result for sufficiently large N . $\square$

Call the simultaneous event in the lemma *balance*. In a fixed epoch, at stage $u \in \{2r, 2r+1\}$, any specified block occurs as one side of at most

$$\Lambda_u = C_{\text{bal}} \left(\frac{m}{2^r} + \log N \right) \quad (18)$$

test frames. Indeed, all tests at that stage belong to one cohort, and each frame side contains the corresponding marked endpoint. Its depth is at least r by (10), so (14) gives the asserted bound.

Although the searches are adaptive, their frames at each stage are the blocks containing the cells' fixed marked endpoints. One occupancy event therefore controls every frame encountered during these searches. Balance depends on the hidden edges; its purpose is to guarantee success of the encoding construction, whose output the strategy will be able to verify directly.

5 Combining the tests of one epoch

Fix an epoch whose earlier answers have been decoded correctly. Its tests and their frames are now known. Write $x_1, \dots, x_R \in \{0, 1\}$ for the answers to its non-forced tests; there are at most m tests at each stage, so $R \leq mD$. We want to recover these bits from only $T = O(m)$ signed sums.

Suppose a matrix $A \in \{-1, 0, 1\}^{T \times R}$ specifies these sums. For each row i , let E_i^+ and E_i^- be the unions of the supports whose entries in that row are $+1$ and -1 , respectively. The supports lie in distinct, edge-disjoint cells. Thus, if their union is a measurable partial matching, two queries give

$$y_i = |E_i^+ \cap M^*| - |E_i^- \cap M^*| = \sum_c A_{ic} x_c. \quad (19)$$

Thus we need two properties: the rows must be realizable by measurements, and the map $x \mapsto Ax$ must be injective on $\{0, 1\}^R$. We call a matrix with the second property *detecting*. Its columns need not be linearly independent (Fig. 5). Geometry determines which sums we may measure; the detecting property determines whether those sums reveal the individual answers.

(a) Four bits, three signed sums

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 1 & -1 \end{pmatrix}$$

$$\ker A = \{t(1, 1, 1, 2) : t \in \mathbb{R}\}.$$

No nonzero vector in this line has all its coordinates in $\{-1, 0, 1\}$.

(b) Two scores give each signed sum

For $x = (1, 0, 1, 1)$:

	positive score	negative score	difference
y_1	$x_1 = 1$	$x_2 = 0$	1
y_2	$x_2 = 0$	$x_3 = 1$	-1
y_3	$x_1 + x_3 = 2$	$x_4 = 1$	1

Figure 5: A detecting matrix can have linearly dependent columns: these four columns give distinct images for all 16 binary inputs. Each signed sum is obtained by subtracting two measured scores.

5.1 Repetition and placement

Some growth in the repetition counts is unavoidable. If every bit appears at most w times, and row i contains r_i copies, the number of possible answer vectors is at most

$$\prod_{i=1}^T (r_i + 1) \leq \left(1 + \frac{wR}{T}\right)^T.$$

A signed sum of r_i bits has at most $r_i + 1$ values, and $\sum_i r_i \leq wR$, so the inequality follows from the arithmetic–geometric mean inequality. For bounded w and $R/T \rightarrow \infty$, this is $2^{o(R)}$, fewer than the 2^R inputs to be distinguished. The obstruction concerns encodings that separate all binary vectors, as we require here.

This does not prescribe a particular repetition schedule. We use geometric growth: it supplies the counting argument with enough repetitions, while the shrinking supports make them possible to place. Over two stages, the relevant block sizes roughly halve. We let the repetition count grow by the smaller factor $\sqrt{2}$:

$$a = 2^{1/4}, \quad w_u = \lceil \alpha a^u \rceil, \quad T = \lceil C_{\text{row}} m \rceil. \quad (20)$$

The positive integer α and the constant C_{row} will be chosen after the detecting theorem has been stated.

Make w_u labeled copies, or *occurrences*, of each non-forced stage- u test. Fix their order in advance, with smaller stages first. A row is *admissible* for an occurrence if

- (i) its left frame block is disjoint from all left frame blocks already in the row;
- (ii) the same holds for its right frame block;
- (iii) the total support size already in the row is at most $\lfloor N/3 \rfloor$.

The first two rules guarantee that the supports form a matching. The last leaves room for one more support: ignoring rounding,

$$\underbrace{\frac{N}{3}}_{\text{current load}} + \underbrace{\frac{N}{2}}_{\text{largest new support}} = \frac{5N}{6} < \underbrace{\frac{7N}{8}}_{\text{measurement capacity}}.$$

All three rules depend only on the known tests and the placements already made. For the existence proof, choose each occurrence's row uniformly from its admissible list; the strategy will instead search over placements.

Lemma 5.1 (Routing with many choices). *Every routing obeying the admissibility rules puts different occurrences of a test in distinct rows and makes each row a partial matching of size at most*

$$\lfloor N/3 \rfloor + \lceil N/2 \rceil \leq 7N/8 \leq N - 2k.$$

For each fixed α , if C_{row} is sufficiently large, then on the balance event every partial routing has at least $T/4$ admissible choices for its next occurrence.

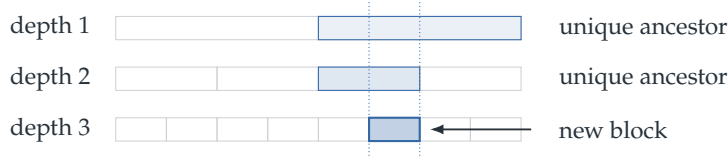
Proof. Two occurrences of one test have the same nonempty frame blocks, so the first two rules prevent them from sharing a row. Those rules also make the supports in each row disjoint on both sides. Before the last insertion, the load is at most $\lfloor N/3 \rfloor$, and the insertion adds at most $\lceil N/2 \rceil$ edges. Thus, for sufficiently large N ,

$$\text{row size} \leq \lfloor N/3 \rfloor + \lceil N/2 \rceil \leq 7N/8 \leq N - 2k. \quad (21)$$

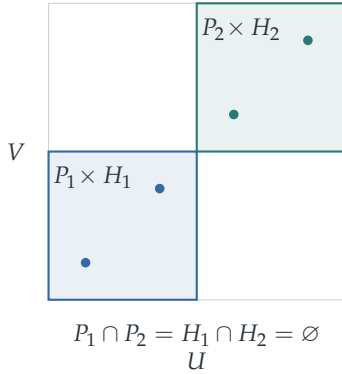
These conclusions do not use balance.

Now assume balance, and consider the next occurrence, at stage v . An earlier occurrence has stage $u \leq v$. On each side, its block is no deeper than the new block. Since all cells use the same hierarchies, an earlier block that meets the new block must be its unique ancestor at that depth (Fig. 6).

Only ancestors can conflict



Admissible frames



Forbidden: a shared V-block

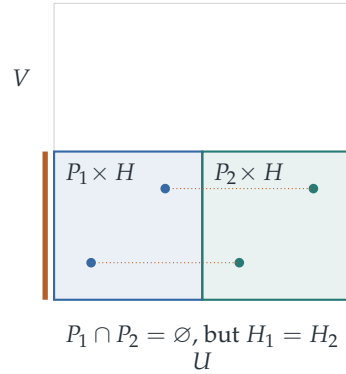


Figure 6: Earlier blocks can meet a new block only by containing it. On each vertex side there is just one possible ancestor at each depth. Our placement rule requires disjoint projections on both sides; this guarantees that the supports form a matching.

At stage u , balance bounds the number of tests using this ancestor by Λ_u from (18). They contribute at most $w_u \Lambda_u$ occurrences. Counting both sides, the number of rows excluded by frame intersections is at most

$$2 \sum_{u \leq v} w_u \Lambda_u. \quad (22)$$

Enlarge this numerical sum to all stages, group stages $2r$ and $2r + 1$, and use $a^2/2 < 1$ to obtain

$$\begin{aligned}
\sum_{u \leq v} w_u \Lambda_u &\leq \sum_{u < D} w_u \Lambda_u \\
&\leq K_\alpha \left[m \sum_{r \geq 0} \left(\frac{a^2}{2} \right)^r + \log N \sum_{u < D} a^u \right] \\
&= O_\alpha(m + a^D \log N) = O_\alpha(m).
\end{aligned} \tag{23}$$

Here K_α depends only on α , and the last step uses $a^D = O(\sqrt{N})$ and $\sqrt{N} \log N = o(m)$. Thus frame intersections exclude $O_\alpha(m)$ rows, regardless of the earlier choices.

To bound the rows excluded by load, use the support-size bound (11) and the fact that there are at most m tests per stage. The total support size of all occurrences is at most

$$\begin{aligned}
\mathcal{V} &\leq m \sum_{r < d} (w_{2r} + w_{2r+1}) s_r \\
&\leq K_\alpha m \left[N \sum_{r \geq 0} \left(\frac{a^2}{2} \right)^r + \sum_{r < d} a^{2r} \right] = O_\alpha(mN).
\end{aligned} \tag{24}$$

At any point, at most $3\mathcal{V}/N = O_\alpha(m)$ rows have load greater than $N/3$. Loads are integers, so these are exactly the rows excluded by the third rule.

In total, at most $K'_\alpha m$ rows are forbidden. Choosing C_{row} large enough that $K'_\alpha m \leq 3T/4$ leaves at least $T/4$ admissible rows. $\square$

The same estimates work for any fixed $1 < a < \sqrt{2}$. The geometric series still converges, and $a^D = O(N^{2 \log_2 a})$ still gives $a^D \log N = o(m)$. The choice $a = 2^{1/4}$ gives the convenient bound $a^D = O(\sqrt{N})$.

Balance has supplied many placement choices at every step. It has not made every completed routing detecting. We next use the freedom in those choices, together with independent signs, to obtain injectivity.

5.2 The detecting theorem

In the following statement, levels represent the stages of an epoch, columns its non-forced tests, and occurrences their labeled copies. The geometry enters only through the number of rows available after each placement history. We will use the theorem to finish the strategy before proving it in [Section 7](#).

Theorem 5.2 (Detecting matrices with adaptive placement constraints). Fix $a > 1$, $0 < \eta \leq 1$, and $K > 0$. There is a positive integer $\alpha = \alpha(a, \eta)$ such that, for every sufficiently large constant $C = C(a, \eta, \alpha, K)$, the following holds for all sufficiently large m .

Consider levels $u = 0, \dots, D-1$, where $D \leq K \log m$, with at most m columns at each level and $R \leq mD$ columns in total. Give each level- u column $w_u = \lceil \alpha a^u \rceil$ labeled occurrences, and fix their order before routing begins. Route them in that order to $T = \lceil Cm \rceil$ rows, subject to these conditions:

- (i) each available list is determined by the routing history so far and contains at least ηT rows;
- (ii) each occurrence is assigned uniformly to a row in its list;
- (iii) occurrences of the same column occupy distinct rows.

Give all occurrences independent uniform signs in $\{-1, 1\}$, independent also of the routing. Let $A \in \{-1, 0, 1\}^{T \times R}$ record the sign of column c in row i , or zero if that column does not occur there. Then A is detecting with probability at least $1/2$.

We now choose the constants for the application. Since $m = \Theta(N / \log N)$, there is an absolute K_{app} such that $D \leq K_{\text{app}} \log m$. Apply Theorem 5.2 with $(a, \eta, K) = (2^{1/4}, 1/4, K_{\text{app}})$ and choose its integer α . Then take C_{row} large enough for both that theorem and Lemma 5.1. Choose N_0 last, so that all the estimates hold. In particular,

$$D \leq K_{\text{app}} \log m, \quad \max_{u < D} w_u = O(\sqrt{N}) = o(m) = o(T). \quad (25)$$

Even the largest repetition count is smaller than the number of rows; the routing rules ensure that its copies actually use distinct rows.

6 Verification and completion of the strategy

The strategy does not test whether the hierarchies are balanced. It tests whether the known supports admit a verified encoding. Balance guarantees that the search succeeds; verification guarantees that every encoding actually used is correct. The finite search precedes all queries in an epoch, so exhausting it costs computation but no queries (Fig. 7).

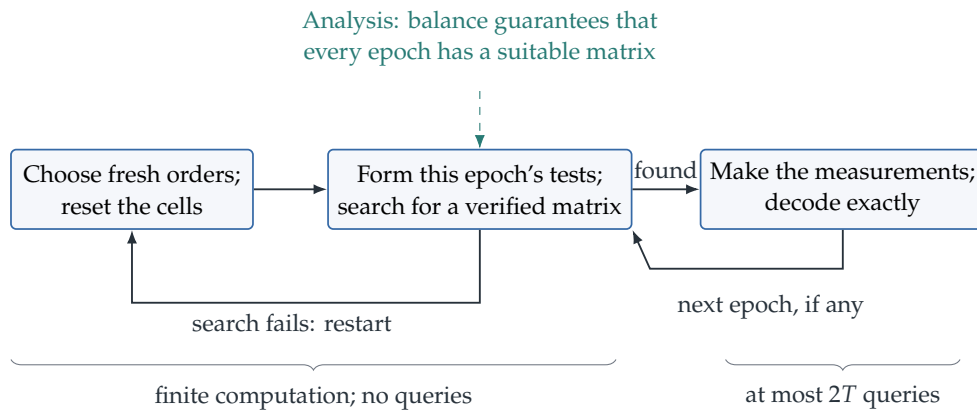


Figure 7: Balance guarantees that the finite search finds an encoding; the strategy checks the encoding itself. An unsuccessful search triggers fresh hierarchies. Only the measurements spend queries.

Lemma 6.1 (One epoch). Assume $\Delta(G) \leq k \leq N/16$ and keep F fixed. Given the known supports and frames of an epoch's tests, there is a procedure that either recovers all their answers exactly with $O(m)$ queries or reports failure before issuing any query. On the balance event, the procedure always recovers the answers.

Proof. Enumerate all complete routings obeying the admissibility rules. For each routing, enumerate all signings and check whether the resulting matrix A is injective on $\{0, 1\}^R$. For example, enumerate the bit vectors and check that their images are distinct. The search is finite and uses only known information. If it finds no detecting matrix, report failure.

Suppose first that balance holds. By [Lemma 5.1](#), every partial routing has at least $T/4$ admissible rows. Uniform choices from these lists, followed by independent uniform signs, satisfy [Theorem 5.2](#) with $\eta = 1/4$. With probability at least $1/2$ the resulting matrix is injective. There is therefore at least one successful routing and signing in the finite search.

Any verified matrix found by the search has measurable rows, by the unconditional part of [Lemma 5.1](#). Two exact measurements per row give $y = Ax$ by (19). Enumerating the bit vectors finds the unique $x \in \{0, 1\}^R$ with this image. It is the true vector of test answers. The query cost is at most $2T = O(m)$; forced answers require no queries. All verification precedes these measurements, so a failed search has incurred no query cost in the epoch. $\square$

Lemma 6.2 (Linear-cost localization). *Assume $\Delta(G) \leq k \leq N/16$ and keep F fixed. The marked edges of all $L \geq \lambda_0 N$ cells from [Lemma 3.1](#) can be identified with zero error using $O(N)$ queries in expectation.*

Proof. An attempt begins by drawing fresh independent hierarchies and placing every cell at its root rectangle $U \times V$. Run the $J + D - 1$ epochs in order, using [Lemma 6.1](#) to decode each one. If an epoch reports failure, abandon the attempt and begin again. Keep the original cells, cohort partition, and set F throughout. Measurements made during an abandoned attempt do not change the fact that each cell has exactly one marked edge.

Every decoded answer is correct, and every forced update follows from (7). Hence the search rectangles always contain their marked edges. Any attempt that finishes all epochs identifies those edges, since both sides of every final rectangle are singletons.

It remains to bound the cost of restarts. Conditional on the entire past, the cells and their marked edges are fixed, while the new hierarchies are independent. By [Lemma 4.1](#),

$$\mathbb{P}(\text{the new attempt is balanced} \mid \text{the past}) \geq 1 - N^{-10}. \quad (26)$$

A balanced attempt cannot fail, by [Lemma 6.1](#). Thus the number of attempts is dominated by a geometric random variable with success probability $1 - N^{-10}$ and has bounded expectation.

Each attempt uses at most $2T(J + D - 1)$ queries. The definitions of m and J give

$$mJ = m \lceil L/m \rceil \leq L + m, \quad mD \leq L.$$

Consequently its cost is $O(m(J + D)) = O(L) = O(N)$. Multiplying by the bounded expected number of attempts proves the query bound, and the correctness argument above proves zero error. $\square$

Lemma 6.3 (One phase). *On an N -by- N residual board, with $N \geq N_0$, one phase identifies at least $\lambda_0 N$ distinct secret edges using $O(N)$ queries in expectation, uniformly over the residual secret.*

Proof. Construct the false-edge reservoir by [Lemma 2.1](#), and fix its sets F and G . Obtain the cells by [Lemma 3.1](#), then locate their marked edges by [Lemma 6.2](#). Each step has expected cost $O(N)$, uniformly over the fixed secret and the history preceding the step. The identified edges are distinct, and there are at least $\lambda_0 N$ of them. $\square$

Proof of Theorem 1.1. Run a phase on the current residual board and delete the endpoints of the identified edges. The next board size satisfies

$$N' \leq (1 - \lambda_0)N. \quad (27)$$

Begin each new phase with fresh randomness and a new false-edge reservoir. The old degree bound need not be small relative to the new board, which is why we rebuild the reservoir. Every residual query is lifted to the original board by adjoining the secret edges already identified.

Let N_r and Q_r be the board size and query cost of phase r , numbered from zero. Set both to zero once the residual board has size less than N_0 , and let $\mathcal{F}_r$ record the history at the start of phase r . The size bound holds for every history, so $N_r \leq n(1 - \lambda_0)^r$. The uniform estimate in Lemma 6.3 gives

$$\mathbb{E}(Q_r \mid \mathcal{F}_r) \leq C_1 N_r$$

for an absolute constant C_1 . Therefore

$$\mathbb{E} \sum_{r \geq 0} Q_r \leq C_1 \sum_{r \geq 0} \mathbb{E} N_r \leq C_1 n \sum_{r \geq 0} (1 - \lambda_0)^r = \frac{C_1}{\lambda_0} n. \quad (28)$$

Solving the final board exhaustively adds only an absolute constant.

Each restart loop has a uniformly positive conditional probability of success, so it terminates almost surely. The board reaches the finite base range after $O(\log n)$ phases. All identified edges follow from exact measurements and verified decoding, and every query is a permutation. This proves the linear expected bound for a zero-error strategy; write Q for its total query count.

We finish by imposing the worst-case bound. Let $Q_{\det}(n) = O(n \log n)$ be a deterministic query bound for the strategy of [EGSS18], and put $q_{\max} = n \lceil \log_2 n \rceil$ for $n \geq 2$. Before our strategy would issue its $(q_{\max} + 1)$ st query, switch to the deterministic strategy on the original board. Markov's inequality gives $\mathbb{P}(Q > q_{\max}) \leq \mathbb{E}Q / q_{\max} = O(1 / \log n)$.

There is also a purely computational restart to limit: an unsuccessful hierarchy attempt may use no queries. Allow at most n hierarchy attempts in each phase, and use the same deterministic fallback if all fail. Each fresh attempt fails with conditional probability at most $1/2$, by (26). There are at most $O(\log n)$ phases, so the probability of any such exhaustion is at most $O(\log n)2^{-n}$.

The modified strategy uses at most $q_{\max} + Q_{\det}(n) = O(n \log n)$ queries on every execution. Its expected query count is at most

$$\mathbb{E}Q + Q_{\det}(n) \left(\frac{\mathbb{E}Q}{q_{\max}} + O(\log n)2^{-n} \right) = O(n).$$

The two cutoffs bound all restart loops, and every intervening computation is finite. Thus the modified strategy terminates with the correct secret for every sequence of random choices and has both claimed bounds. $\square$

7 Why the matrix is detecting

We now work in the setting of Theorem 5.2. There are at most m columns at each level $u < D$, each with $w_u = \lceil \alpha a^u \rceil$ occurrences, and $T = \lceil Cm \rceil$ rows. The available rows may depend on the previous placements, but every list contains at least ηT rows; occurrences of one column occupy distinct rows, and the signs are independent of the routing.

Two distinct binary vectors have the same image precisely when their nonzero difference lies in the kernel. We must therefore exclude every nonzero vector in $\{-1, 0, 1\}^R$ from the kernel of A . Fix such a vector z , and say that it *selects* the columns where it is nonzero. A selected column contributes all its occurrences. If k_u selected columns lie at level u , put

$$W(z) = \sum_{u < D} k_u w_u. \quad (29)$$

We call this the *occurrence weight* of z . Counting selected columns alone would ignore their levels: a deep column contributes many more occurrences than a shallow one, and all those occurrences must cancel. The same weights that make cancellation less likely also restrict how many difference vectors can have a given weight.

Absorb each nonzero coordinate z_c into the signs of its occurrences. The resulting signs are still independent and uniform in $\{-1, 1\}$, independently of the routing. The equation $Az = 0$ says that their sum in every row is zero.

We will need two different estimates for this event. At small weights, cancellation forces the selected occurrences to meet in pairs, and a prescribed encounter has probability $O(1/T)$. This dependence on T is essential: at a fixed weight there may already be polynomially many candidate vectors, so a bound merely exponential in the weight would not suffice. At large weights, encounters are plentiful. We instead use the independent signs, showing that cancellation becomes less likely as the load on the rows increases. A bound based only on the number of occupied rows would lose this further gain.

7.1 Cancellation of a fixed vector

Lemma 7.1 (Sparse pairing bound). *Under the hypotheses of [Theorem 5.2](#), there is a constant K_η , depending only on η , such that every nonzero z of weight W satisfies*

$$\mathbb{P}(Az = 0) \leq \left(\frac{K_\eta W}{T} \right)^{W/2}. \quad (30)$$

If W is odd, this probability is zero.

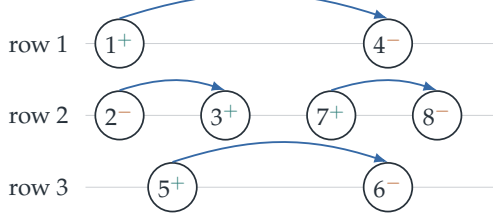
Proof. If every row has sign sum zero, its selected occurrences can be paired with opposite signs. Thus W is even, and there is a perfect pairing whose members share a row and have opposite signs.

Fix a pairing and orient each pair from its earlier to its later member in the prescribed routing order. When the later member is routed, the earlier member's row is already known ([Fig. 8](#)). Conditional on the complete past, the chance of hitting that row is at most $1/(\eta T)$. Apply this bound at each later member. Even when pairs interleave, every constraint imposed earlier is determined by the current history, so successive conditioning bounds the probability that all pairs share rows by $(\eta T)^{-W/2}$.

Now expose the signs. Independently of the routing, the two signs in each pair are opposite with probability $1/2$, and these events are independent across pairs. The fixed pairing therefore meets both requirements with probability at most $(2\eta T)^{-W/2}$. There are $(W-1)!! \leq W^{W/2}$ pairings. A union bound proves the claim with $K_\eta = (2\eta)^{-1}$. $\square$

Cancellation forces pairs

Order labels; signs as superscripts.



One row constraint per pair

When the later member is placed, the earlier member's row is already known.

$$\mathbb{P}(\text{specified row} \mid \text{past}) \leq \frac{1}{\eta T}.$$

For a fixed pairing of W occurrences:

$$\underbrace{(\eta T)^{-W/2}}_{\text{common rows}} \underbrace{2^{-W/2}}_{\text{opposite signs}}.$$

Figure 8: Cancellation pairs the selected occurrences within rows with opposite signs. In a prescribed pairing, the later member must hit the earlier member's row, imposing one conditional placement constraint per pair.

To use the signs at larger weights, we first show that a block of selected occurrences usually reaches many different rows. The available-list condition gives this conclusion after every routing history; independence of the row choices is unnecessary.

Lemma 7.2 (Adapted distinct-row bound). *There are constants $\rho, \gamma > 0$, depending only on η , with the following property. Fix in advance a block of $1 \leq s \leq T$ selected occurrences in their induced chronological order; unselected occurrences may be interspersed. Conditional on any routing history before the first occurrence of the block,*

$$\mathbb{P}(\text{the selected block visits fewer than } \rho s \text{ distinct rows}) \leq e^{-\gamma s}. \quad (31)$$

Proof. Set $\rho = \eta/16$, and condition on any history preceding the block. Until the block has visited ρs distinct rows, the conditional probability that its next selected occurrence revisits a row is at most

$$\frac{\rho s}{\eta T} \leq \frac{1}{16}.$$

After the threshold is reached, set all subsequent revisit indicators to zero.

If the block visits fewer than ρs rows, at least $q = \lceil s/2 \rceil$ of its occurrences are revisits. For any prescribed set of q occurrence indices, successive conditioning on the complete routing history bounds the probability that all their indicators are one by 16^{-q} . A union bound gives

$$\mathbb{P}(\text{fewer than } \rho s \text{ rows are visited}) \leq \binom{s}{q} 16^{-q} \leq 2^s 16^{-s/2} = 2^{-s}.$$

Thus we may take $\gamma = \log 2$. □

Lemma 7.3 (Anticoncentration for a fixed vector). *There are constants $c_1, c_2 > 0$, depending only on η , such that, for all sufficiently large T , every nonzero z of weight W satisfies*

$$\mathbb{P}(Az = 0) \leq \begin{cases} 2e^{-c_1 W}, & 1 \leq W \leq T, \\ e^{-c_2 W} + \exp\left[-c_2 T \log\left(1 + \frac{W}{T}\right)\right], & W \geq T. \end{cases} \quad (32)$$

Proof. Fix the routing. A row with $\ell > 0$ selected occurrences balances with probability zero when ℓ is odd. When ℓ is even, the central-binomial estimate gives

$$2^{-\ell} \binom{\ell}{\ell/2} \leq (1 + \ell)^{-1/2} \leq 2^{-1/2}.^2 \quad (33)$$

The signs in different rows are independent, so these probabilities multiply.

Suppose first that $W \leq T$. By Lemma 7.2, the selected occurrences visit at least ρW rows, except with probability $e^{-\gamma W}$. Conditional on such a routing, the probability that all these rows balance is at most $2^{-\rho W/2}$. Adding the two bounds proves the first line of (32), with c_1 sufficiently small.

Now suppose that $W \geq T$, and put $q = \lfloor W/T \rfloor \geq 1$. Divide the first qT selected occurrences into q consecutive blocks of T selected occurrences, leaving aside the fewer than T that remain. Call a block *good* if it visits at least ρT rows. The conditional form of Lemma 7.2 shows that any prescribed set of s blocks are all bad with probability at most $e^{-\gamma Ts}$: the estimate applies to each specified block after every possible history preceding it. Hence

$$\mathbb{P}(\text{at least } q/2 \text{ blocks are bad}) \leq 2^q e^{-\gamma T q/2} \leq e^{-cTq} \leq e^{-c'W} \quad (34)$$

for sufficiently large T , where the last inequality uses $W < (q+1)T \leq 2qT$.

On the complementary event, more than half the blocks are good. The total load alone would not prevent most occurrences from accumulating in a few rows. Instead, count how many good blocks visit each row. Every good block visits many distinct rows, while any one row is counted at most once per block. These capped counts will force a sufficiently small product of the row probabilities.

For each row i , let g_i count the good blocks that visit it, and let ℓ_i be its total selected load, including the remainder. Then

$$\sum_i g_i \geq \frac{\rho T q}{2}, \quad 0 \leq g_i \leq q, \quad \ell_i \geq g_i. \quad (35)$$

A visit from a new block certifies a further occurrence, irrespective of repeated visits from that block (Fig. 9). Concavity of $\log(1+x)$ gives the chord inequality

$$\log(1+g) \geq \frac{g}{q} \log(1+q) \quad (0 \leq g \leq q).$$

It follows that

$$\begin{aligned} \sum_i \log(1+\ell_i) &\geq \sum_i \log(1+g_i) \\ &\geq \frac{\log(1+q)}{q} \sum_i g_i \geq \frac{\rho T}{2} \log(1+q). \end{aligned} \quad (36)$$

Multiplying the first bound in (33) over the occupied rows, with a factor 1 for each empty row, now gives

$$\mathbb{P}(Az = 0 \mid \text{routing}) \leq \exp \left(-\frac{1}{2} \sum_i \log(1+\ell_i) \right) \leq \exp(-cT \log(1+q)).$$

Since $q = \lfloor W/T \rfloor \geq 1$, the quantities $\log(1+q)$ and $\log(1+W/T)$ differ by at most an absolute constant factor. Combining this estimate with (34) proves the second line of (32), after decreasing c_2 if necessary. $\square$

²For completeness, put $p_j = 4^{-j} \binom{2j}{j}$. The estimate $p_j \leq (2j+1)^{-1/2}$ holds at $j=1$, and induction follows from $p_{j+1} = p_j(2j+1)/(2j+2) \leq \sqrt{2j+1}/(2j+2) \leq (2j+3)^{-1/2}$.

Block visits force row load

	good blocks				
	G_1	G_2	G_3	g_i	ℓ_i
row 1	2	0	1	2	≥ 3
row 2	1	2	0	2	≥ 3
row 3	2	1	1	3	≥ 4
row 4	1	1	2	3	≥ 4
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

The row bounds multiply

A repeated visit increases ℓ_i , but contributes only once to g_i . Hence $g_i \leq \ell_i$.

$$\begin{aligned} \mathbb{P}(Az = 0 \mid \text{routing}) \\ \leq \prod_i (1 + \ell_i)^{-1/2}. \end{aligned}$$

Figure 9: Each good block visits many rows, but contributes at most one to each count g_i . These counts give lower bounds on the row loads ℓ_i that can be combined through the chord inequality for $\log(1 + g)$.

7.2 Counting by occurrence weight

We have bounded cancellation for each fixed difference vector. We now count those vectors on the same scale. Selecting a column at level u incurs weight w_u and allows two choices of sign; leaving it unselected incurs no weight. Keeping these costs in the count will be much more effective than using the total 3^R .

Lemma 7.4 (Counting difference vectors by weight). *Let*

$$\mathcal{N}(S) = \#\{z \in \{-1, 0, 1\}^R : W(z) \leq S\}.$$

There is a constant K_a , depending only on a , such that

$$\log \mathcal{N}(S) \leq \begin{cases} K_a \frac{S}{\alpha} \log \frac{e\alpha m}{S}, & 0 < S \leq \alpha m, \\ K_a m \left(1 + \log \frac{S}{\alpha m}\right), & S \geq \alpha m. \end{cases} \quad (37)$$

Proof. For $\lambda > 0$, each vector of weight at most S contributes at least $e^{-\lambda S}$ to $\sum_z e^{-\lambda W(z)}$. This sum factors over coordinates: a level- u coordinate contributes $1 + 2e^{-\lambda w_u}$. Padding each level to m coordinates can only increase the sum, so

$$\mathcal{N}(S) \leq e^{\lambda S} \sum_z e^{-\lambda W(z)} \leq e^{\lambda S} \prod_{u < D} (1 + 2e^{-\lambda w_u})^m.$$

Using $w_u \geq \alpha a^u$ and $\log(1 + x) \leq x$, we obtain

$$\log \mathcal{N}(S) \leq \lambda S + 2m \sum_{u \geq 0} e^{-\lambda \alpha a^u}. \quad (38)$$

Increasing λ suppresses the sum but enlarges the first term. We choose it differently on the two sides of $S = \alpha m$.

For $S \leq \alpha m$, choose

$$\lambda = \alpha^{-1} \log(e\alpha m/S), \quad t = \lambda \alpha \geq 1.$$

This makes $e^{-\lambda \alpha} = S/(e\alpha m)$, so the lowest level contributes only $O(S/\alpha)$ to the second term of (38). The higher levels together cost only a constant factor more: since $a^u \geq 1 + (a-1)u$,

$$\sum_{u \geq 0} e^{-ta^u} \leq e^{-t} \sum_{u \geq 0} e^{-t(a-1)u} \leq K'_a e^{-t}.$$

Thus (38) is at most $(S/\alpha) \log(e\alpha m/S) + O_a(S/\alpha)$, which proves the first bound because $t \geq 1$.

For $S \geq \alpha m$, write $q = S/(\alpha m) \geq 1$ and choose $\lambda = m/S = 1/(\alpha q)$. The first term in (38) is now exactly m . In the sum, the transition occurs where αa^u is of order S/m . More precisely, split at $u_0 = \lceil \log_a q \rceil$. There are $O_a(1 + \log q)$ terms up to this index, each at most 1, and the later terms have a convergent tail bounded by $\sum_{j \geq 0} e^{-a^j}$. Therefore

$$\sum_{u \geq 0} e^{-a^u/q} \leq K_a''(1 + \log q),$$

and substitution into (38) proves the second bound. $\square$

7.3 Completing the union bound

Proof of Theorem 5.2. The two large constants have different roles. The repetition constant α makes the counting cost small relative to the cancellation estimates at low and intermediate weights. Once α is fixed, increasing the row constant C makes the large-weight cancellation estimate dominate the remaining counting cost. All constants below may depend on a and η ; we choose the integer α first, then C , and finally take m sufficiently large.

Every nonzero difference vector has weight at least α . Partition the possible weights into dyadic shells

$$\mathcal{Z}_S = \{z : S \leq W(z) < 2S\}, \quad S = \alpha 2^j \quad (j \geq 0). \quad (39)$$

For each shell, we multiply a bound for a fixed vector by the count $\mathcal{N}(2S)$. The estimates to use change at the two scales αm and T , as recorded in Table 1.

Shell range	Cancellation estimate	Counting estimate
$2S \leq \alpha m$	Sparse pairing	First line of (37)
$\alpha m \leq 2S \leq T$	$2e^{-c_1 W}$	Second line of (37)
$T \leq 2S$	Both lines of (32)	Second line of (37)

Table 1: The three estimates for a shell $S \leq W < 2S$. A shell crossing T requires both fixed-vector bounds.

Sparse shells: $2S \leq \alpha m$. Require $C \geq 2K_\eta \alpha$. For $S \leq W < 2S$, the base $2K_\eta S/T$ is at most one, so

$$\left(\frac{K_\eta W}{T}\right)^{W/2} \leq \left(\frac{2K_\eta S}{T}\right)^{W/2} \leq \left(\frac{2K_\eta S}{T}\right)^{S/2}.$$

By Lemmas 7.1 and 7.4,

$$\log \mathbb{P}(\exists z \in \mathcal{Z}_S : Az = 0) \leq \frac{2K_\alpha S}{\alpha} \log \frac{e\alpha m}{2S} - \frac{S}{2} \log \frac{T}{2K_\eta S}. \quad (40)$$

Put $\xi = \alpha m/S \geq 2$. Choose α so that $2K_\alpha/\alpha < 1/8$, and then take C/α sufficiently large. Since

$$\log(e\xi/2) \leq 1 + \log \xi, \quad \frac{T}{2K_\eta S} \geq \frac{C}{2K_\eta \alpha} \xi,$$

the right-hand side of (40) is at most

$$-cS(1 + \log \xi) \quad (41)$$

for an absolute $c > 0$. Write $S = \alpha t$, where $t = 2^j$. The function $t(1 + \log(m/t))$ increases on $1 \leq t \leq m$, so the weakest decay occurs at $t = 1$. The sum over the sparse shells is therefore at most

$$O(\log m) \exp[-c\alpha(1 + \log m)] = o(1).$$

Intermediate shells: $S \geq \alpha m/2$ and $2S \leq T$. The first line of (32) and the second line of (37) give

$$\log \mathbb{P}(\exists z \in \mathcal{Z}_S : Az = 0) \leq \log 2 + K_a m \left(1 + \log \frac{2S}{\alpha m} \right) - c_1 S. \quad (42)$$

For $\zeta = S/(\alpha m) \geq 1/2$, the ratio $(1 + \log(2\zeta))/\zeta$ is bounded. Taking α sufficiently large, still before choosing C , makes (42) at most $-c'S$ for sufficiently large m . There are $O(\log m)$ such shells, each with $S = \Omega(m)$, so their total contribution is $o(1)$.

Large and transition shells: $S \geq T/2$. Require also $C \geq \alpha$. A shell in this range may cross T . For its vectors with $W < T$, use the first line of (32); for those with $W \geq T$, use the second. After decreasing c_2 , both cases satisfy

$$\mathbb{P}(Az = 0) \leq 3e^{-c_2 S} + \exp \left[-c_2 T \log \left(1 + \frac{S}{T} \right) \right]. \quad (43)$$

Since $2S \geq T \geq \alpha m$, the second counting bound applies. Writing $\zeta = S/T \geq 1/2$ and using $T \leq (C+1)m$, we have

$$\log \mathcal{N}(2S) \leq K_a m \left(1 + \log \frac{2(C+1)\zeta}{\alpha} \right). \quad (44)$$

Uniformly for $\zeta \geq 1/2$, the expression in parentheses divided by either ζ or $\log(1 + \zeta)$ is $O(1 + \log C)$, with α now fixed. As $C/(1 + \log C) \rightarrow \infty$, we may choose C so that (44) is at most each of

$$\frac{c_2}{2} C m \zeta \quad \text{and} \quad \frac{c_2}{2} C m \log(1 + \zeta). \quad (45)$$

Because $S = T\zeta \geq C m \zeta$ and $T \geq C m$, multiplying (43) by $\mathcal{N}(2S)$ leaves at most

$$3e^{-c_2 S/2} + e^{-(c_2/2)T \log(1+\zeta)}.$$

Both terms are $e^{-\Omega(m)}$ throughout this range.

The three ranges cover every shell: if $2S > \alpha m$, either $2S \leq T$, giving an intermediate shell, or $S > T/2$, giving a large or transition shell. There are only $O(\log m)$ nonempty shells in all, since

$$W \leq m \sum_{u < D} \lceil \alpha a^u \rceil = m^{O(1)}$$

by $D \leq K \log m$. Summing the bounds shows that the probability of a nonzero ternary kernel vector tends to zero, and is therefore less than $1/2$ for sufficiently large m . Every difference of binary vectors is ternary, so outside this event A is injective on $\{0, 1\}^R$. $\square$

References

- [EGSS18] Mourad El Ouali, Christian Glazik, Volkmar Sauerland, and Anand Srivastav. “On the Query Complexity of Black-Peg AB-Mastermind”. In: *Games* 9.1 (2018). issn: 2073-4336. doi: [10.3390/g9010002](https://doi.org/10.3390/g9010002). url: <https://www.mdpi.com/2073-4336/9/1/2> (pages 1, 15).
- [MS24] Anders Martinsson and Pascal Su. “Mastermind with a Linear Number of Queries”. In: *Combinatorics, Probability and Computing* 33.2 (2024), pp. 143–156. doi: [10.1017/S0963548323000366](https://doi.org/10.1017/S0963548323000366). url: <https://doi.org/10.1017/S0963548323000366> (page 1).
- [Sub26] Bernardo Subercaseaux. “Price of Locality in Permutation Mastermind: Are TikTok Influencers Chaotic Enough?” In: *13th International Conference on Fun with Algorithms, FUN 2026, Porquerolles, France, May 18-22, 2026*. Ed. by John Iacono. Vol. 366. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2026, 39:1–39:21. doi: [10.4230/LIPICS.FUN.2026.39](https://doi.org/10.4230/LIPICS.FUN.2026.39). url: <https://doi.org/10.4230/LIPICS.FUN.2026.39> (page 1).